

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number 13

Start date: 03/02/2026

Name: Onkar Vyawahare

Internship Duration: 8 Weeks

1. Project Title

Interactive Analytics Dashboard and Web Portal for Global Income Distribution.

2. Project Objective

The main objective of my internship project was to analyze global income inequality and present it in a meaningful, interactive, and visually engaging way.

I aimed to transform raw socioeconomic data into actionable insights using data analytics and full-stack development. The project focuses on helping users understand income distribution patterns, inequality severity, and inclusive development across countries.

Additionally, the goal was to build a complete end-to-end solution, including:

- Data collection and preprocessing
- Analytical dashboard development
- Full-stack web application deployment
- User interaction and admin monitoring system.

3. Project description in detail

The project focuses on analyzing and visualizing global income inequality using an integrated data analytics and web-based approach. It is designed to help users understand how income is distributed across different countries and how economic growth relates to social equality.

At its core, the project provides an interactive analytical dashboard that presents key indicators such as income inequality levels, economic performance, and inclusive development. The dashboard enables users to explore trends, compare countries, and identify patterns in income distribution through dynamic visualizations.

To enhance accessibility and usability, the project was extended into a full-stack web application. This platform serves as a centralized system where users can not only view the dashboard but also interact with it in a seamless and user-friendly environment.

A key highlight of the system is the Admin Portal, which provides administrative control over the platform. It allows monitoring of user activity, management of user accounts, and handling of queries submitted through the system. This adds a layer of scalability and real-world applicability to the project.

From a technological perspective, the project integrates data analytics tools with modern web development frameworks, creating a complete end-to-end solution. It demonstrates how raw data can be transformed into meaningful insights and delivered through an interactive digital platform.

In real-world scenarios, this project can be useful for researchers, students, and policymakers who want to study economic inequality and make data-driven decisions. It simplifies complex datasets into intuitive visuals, making analysis more accessible and impactful.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Data collection	Collected datasets from World Bank
Week 2	Data cleaning	Completed data cleaning
Week 3	Data transformation, Power BI setup	Converted and merged datasets, Data validation and setup
Week 4	Measures creation, Dashboard development	Created analytical KPIs, Built interactive dashboard
Week 5	Web Exploration	Gone through and accumulated all the possible designs and features to include
Week 6	Web development	Developed frontend and backend
Week 7	Web Enhancements	Enhanced the website dashboard functionalities
Week 8	Final execution	PPT, demo, and documentation

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Data collection started	03 Feb 2026
Prototype/First Draft	Initial dashboard created	20 Feb 2026
Mid-Term Review	Dashboard finalization and enhancements	27 Feb 2026
Final Submission	Web portal completed	20 Mar 2026
Presentation	PPT and demo completion	26 Mar 2026

5b. Project execution details

This project was developed in multiple structured phases:

Milestone 1 – Data Collection & Preparation

In the first phase, I collected datasets related to global income inequality from the World Bank database. The key indicators included:

- Gini Index
- Income share held
- GDP per capita
- Unemployment Rate

I performed data cleaning such as:

- Removing unnecessary columns
- Standardizing column names
- Handling missing values using interpolation and averaging
- Converting data formats

Finally, I converted all datasets into long format and merged them into a single master dataset.

Milestone 2 – Data Transformation & Dashboard Development

In this phase, I used Power BI to build an analytical dashboard.

I created advanced calculated measures such as:

- GDP YoY Growth %
- Gini Change YoY
- Unemployment YoY Change
- Polarization Index

I also created classification columns to categorize countries based on inequality levels and economic performance.

The dashboard includes:

- KPI summary cards
- Global Overview
- Trend analysis charts
- Comparative analysis sections

This allows users to explore data interactively and gain meaningful insights.

Milestone 3 – Web Portal Development

In this phase, I developed a full-stack web application to host the dashboard.

The platform includes:

- Public pages explaining global inequality
- User login and authentication system
- Embedded Power BI dashboard
- Country comparison tools
- Data download features

An Admin Portal was also developed with:

- User management system
- Session tracking
- Query management
- Analytics monitoring

Technologies used:

- Frontend: HTML5, CSS3, JavaScript
 - Backend: Flask (Python 3.x)
 - Database: SQLite, SQL Alchemy
 - Authentication: Flask-Login
-

Milestone 4 – Final Execution & Presentation

In this phase, our instructor divided us into two groups:

- PPT Presentation Team
- Demo Team

Both groups worked simultaneously on:

- Individual documentation preparation
- Group documentation
- PPT creation
- Website enhancements

During this phase, I contributed to:

- Refining the website features
- Preparing documentation
- Supporting presentation and demo readiness

This phase helped in improving both technical and communication skills.

6. Snapshots / Screenshots

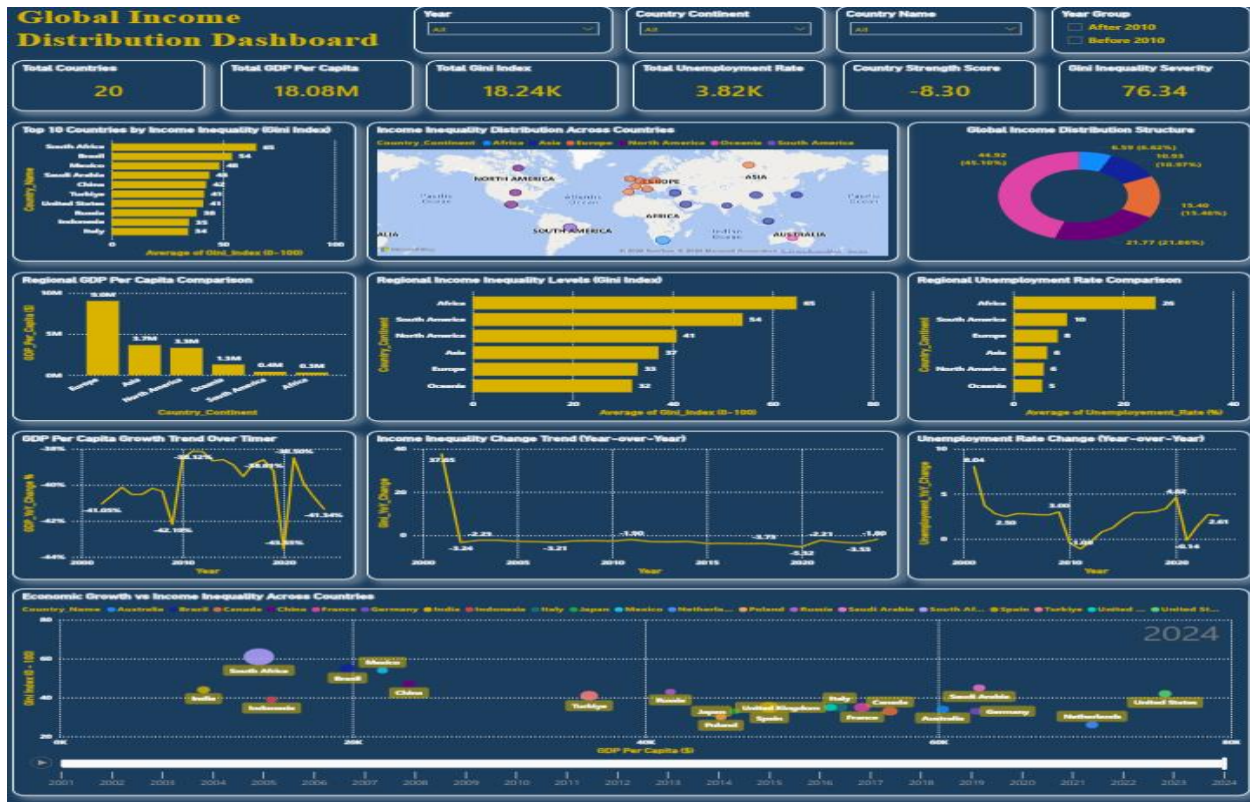
Dataset After Cleaning and Merging

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	Country Name	Country Code	year	GDP_Per_Capita	Gini_Index	Income Share_Lowest 20%	Income Share_Second 20%	Income Share_Third 20%	Income Share_Fourth 20%	Income Share_Highest 20%	Unemployment_Rate			
1	United States	USA	2001	55017.30498	40.6	5.5	10.7	15.6	22.3	45.9	4.731			
2	United States	USA	2002	55436.11698	40.4	5.4	10.7	15.7	22.6	45.6	5.783			
3	United States	USA	2003	56498.20783	40.8	5.3	10.5	15.6	22.7	45.9	5.089			
4	United States	USA	2004	58131.63369	40.5	5.4	10.6	15.6	22.6	45.8	5.529			
5	United States	USA	2005	59604.75364	41.3	5.3	10.5	15.4	22.4	46.5	5.084			
6	United States	USA	2006	60676.56557	41.7	5.3	10.3	15.3	22.3	46.8	4.623			
7	United States	USA	2007	61306.59694	41.1	5.3	10.4	15.5	22.6	46.2	4.622			
8	United States	USA	2008	60796.43384	41.1	5.3	10.4	15.5	22.7	46.2	5.784			
9	United States	USA	2009	58714.97369	40.8	5.5	10.4	15.4	22.5	46.1	9.254			
10	United States	USA	2010	59789.41291	40.2	5.4	10.5	15.6	22.3	45.4	9.633			
11	United States	USA	2011	60245.4844	41.2	5.3	10.3	15.4	22.8	46.3	8.949			
12	United States	USA	2012	61134.53061	41.2	5.3	10.3	15.4	22.8	46.2	8.069			
13	United States	USA	2013	61958.72344	40.9	5.4	10.3	15.4	22.8	46.1	7.375			
14	United States	USA	2014	63018.80339	41.7	5.3	10.2	15	22.5	47	6.168			
15	United States	USA	2015	64359.47259	41.5	5.4	10.2	15.1	22.6	46.7	5.28			
16	United States	USA	2016	65017.65496	41.3	5.3	10.3	15.2	22.7	46.5	4.869			
17	United States	USA	2017	66155.55557	41.4	5.2	10.2	15.3	22.8	46.4	4.355			
18	United States	USA	2018	67719.76564	41.8	5.2	10.1	15.2	22.6	46.9	3.896			
19	United States	USA	2019	69112.60269	41.9	5.2	10.1	15.1	22.6	47	3.669			
20	United States	USA	2020	67342.07953	40	5.7	10.6	15.5	22.7	45.5	8.055			
21	United States	USA	2021	71307.40173	39.7	6.1	10.7	15.3	22.2	45.7	5.349			
22	United States	USA	2022	72679.25784	41.7	5.2	10.2	15.2	22.6	46.8	3.65			
23	United States	USA	2023	74158.71642	41.8	5.1	10.1	15.2	22.7	46.8	3.638			
24	United States	USA	2024	75489.26643	42	3.1	8.5	13.4	22.6	52.2	4.022			
25	China	CHN	2001	4372.157056	45	6.5	10.5	14.4	22.4	47.9	3.8			
26	China	CHN	2002	4744.398551	42	5.7	9.6	14.4	22.4	47.9	4.24			
27	China	CHN	2003	5192.006046	47.9	5.8	10	14.5	22.4	47.5	4.58			
28	China	CHN	2004	5684.168731	47.3	5.8	10	14.6	22.4	47.3	4.49			
29	China	CHN	2005	6298.323185	40.9	5.9	10	14.7	22.3	47.1	4.52			
30	China	CHN	2006	7057.099333	46.7	5.8	10.3	14.7	22.3	47.8	4.43			

After Calculated Columns and Measures creation

Column Name	Type
Country_Code	Column
Country_Continent	Column
Country_Name	Column
Country_Strength_Score	Column
GDP_Per_Capita (\$)	Column
GDP_Per_Capita_Category	Column
GDP_YoY_Change %	Column
Gini_Index (0-100)	Column
Gini_Inequality_Level	Column
Gini_Inequality_Severity_Score	Column
Gini_YoY_Change	Column
Income_Share_Fourth 20%	Column
Income_Share_Highest 20%	Column
Income_Share_Lowest 20%	Column
Income_Share_Second 20%	Column
Income_Share_Third 20%	Column
Income_Polarization_Index	Column
Income_Share_Gap	Column
Middle_Class_Income_Share	Column
Region_GDP_%_Contribution	Column
Region_Total_GDP	Column
Total_Countries	Column
Unemployment_Rate (%)	Column
Unemployment_Region_Average	Column
Unemployment_Level	Column
Unemployment_Rank	Column
Unemployment_YoY_Change	Column
year	Column
Year_Group	Column

Dashboard Overview



Authentication System

GIDA

GLOBAL INCOME DISTRIBUTION ANALYTICS

Register Login

GIDA

GLOBAL INCOME DISTRIBUTION ANALYTICS

USER LOGIN ADMIN LOGIN

CONSUMER ACCESS

Sign in to explore interactive wealth datasets

EMAIL ADDRESS

john@example.com

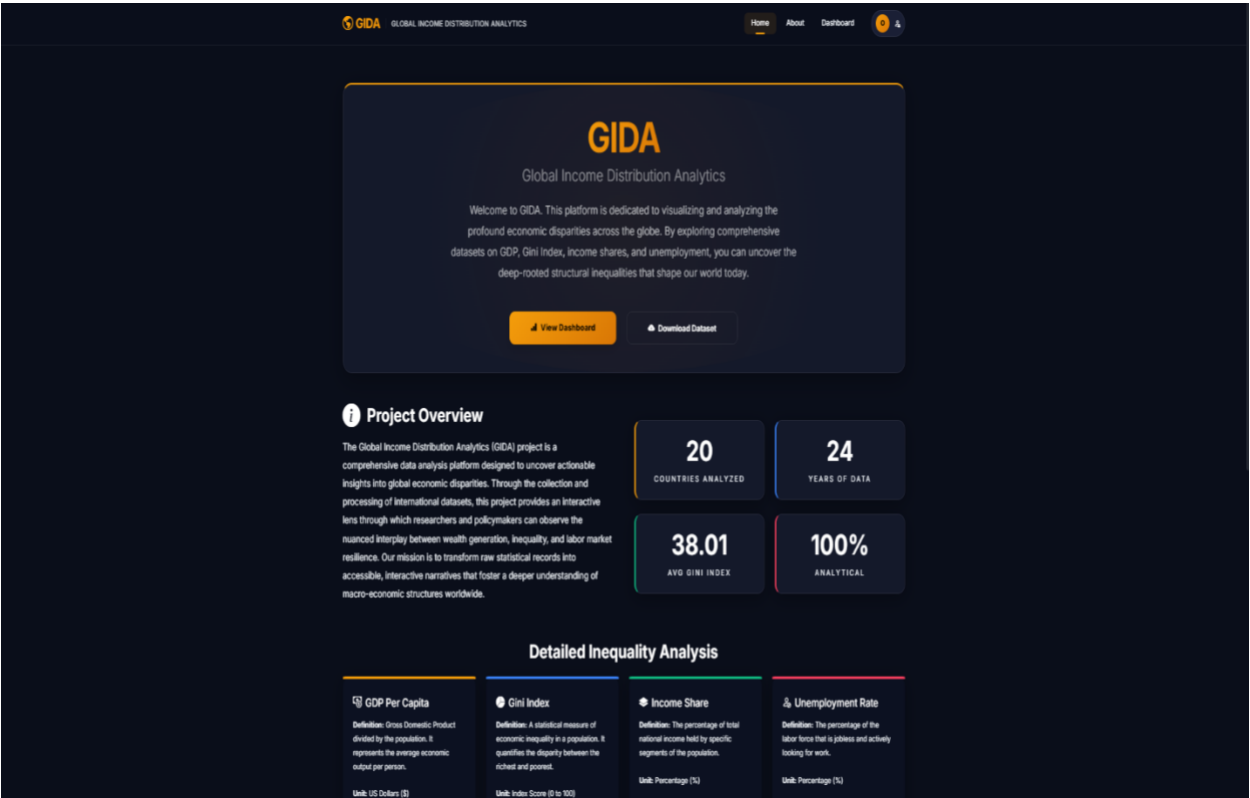
PASSWORD

Sign In as User

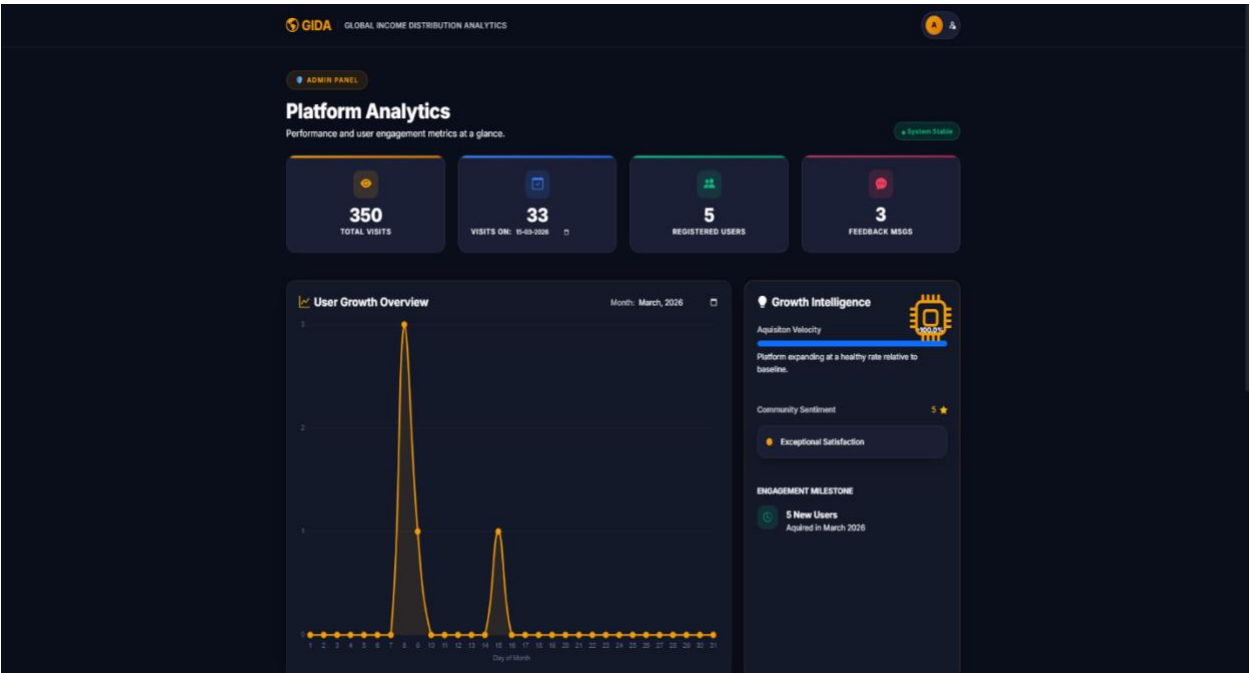
New to GIDA? Create Account

© 2026 Built with Flask & Power BI

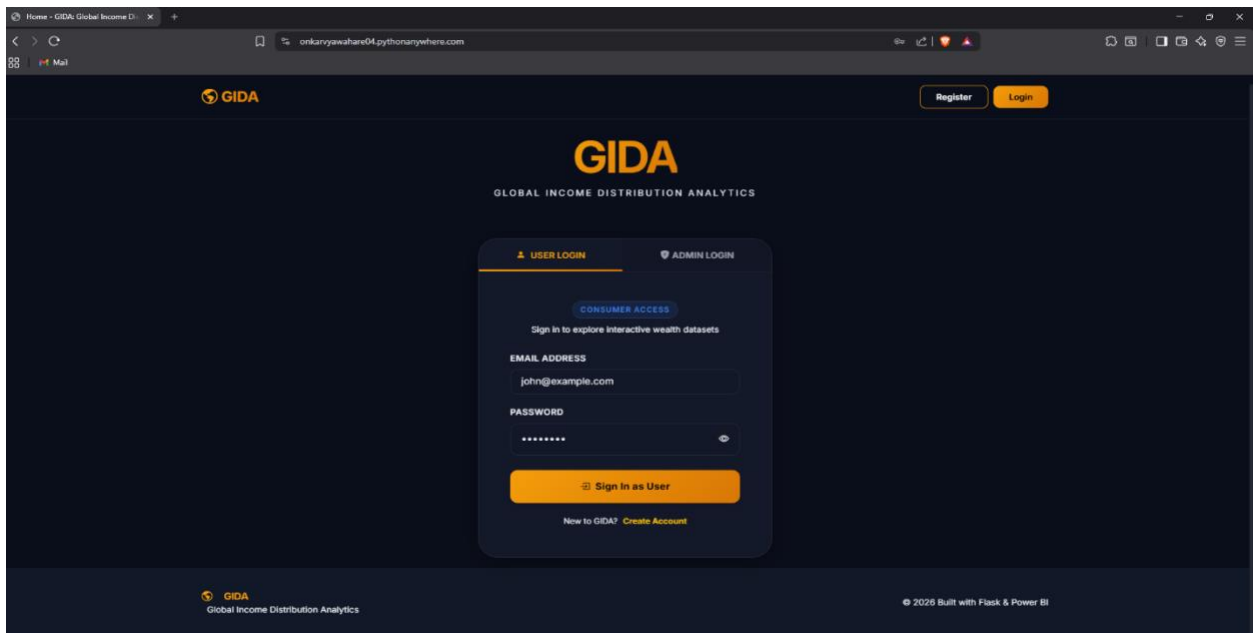
Home Page



Admin Portal



Deployed Website



7. Challenges Faced

Some challenges I faced during the project were:

- Handling missing and inconsistent data
- Creating Insightful KPI Cards
- Integrating Power BI with web application
- Implementing secure authentication
- Managing backend and frontend communication

I resolved these by:

- Applying proper data cleaning techniques
- Learning DAX functions
- Debugging integration issues
- Testing authentication flows.

8. Learnings & Skills Acquired

Technical Skills

- Data Cleaning and Transformation
- Power BI Dashboard Development
- DAX Calculations
- Full-stack Development
- Database Design (SQLite)

Soft Skills

- Problem-solving
- Team collaboration
- Documentation
- Presentation skills.

9. Testimonials from team

This internship gave me hands-on experience in solving real-world problems. I was able to apply my knowledge in data analytics and development effectively.

It helped me improve my confidence in building complete projects and presenting them.

10. Conclusion

This internship was a valuable learning experience.

I successfully built a complete system that analyzes global income inequality using both data analytics and web development.

It strengthened my technical skills and aligned with my career goal of becoming a developer.

11. Acknowledgements

I would like to thank Infosys Springboard for providing this opportunity.

I am grateful to my instructor and mentor for their guidance

I also thank my teammates for their support and collaboration throughout the project.

---- With this the Internship is Completed Successfully ----